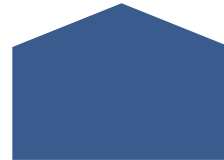
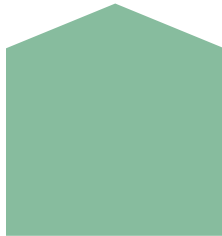
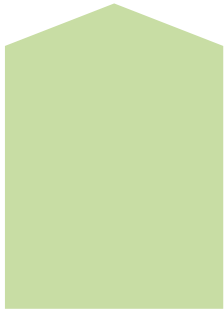


BSCS 506-Software Project Management

LECTURE#02

Course Supervisor: Dr. Humera Azam



- Introduction of SPM Tools,
- PMI's Knowledge Areas,
- Technical Fundamentals in SPM,
- Lifecycle Relationships,
- Classic Mistakes Product-Process-Peoples-Technology Mistakes PMI Framework,
- PMI Process Groups: Process Initiating Process Group,
 - Planning Process Process Group,
 - Executing Process,
 - Process Monitoring and controlling,
 - Closing Process Group,
 - Project Charter,
 - Statement of Work

Introduction of SPM Tools

computer programs are not project management: they are tools(**software**) for software project managers to use: Project management software describes the **tools** to efficiently coordinate and automate the various project management component processes.

Project Management Tools



Project Management tools help to keep a streamline flow in the Project Management process. The tools help the team or an individual to carry out the steps of project management in an effective and efficient manner

**PROJECT
SCHEDULING**

**RESOURCE
MANAGEMENT**

**DOCUMENT
MANAGEMENT**

**TASK
MANAGEMENT**

**TIME
TRACKING**

**TEAM
COLLABORATION**

REPORTING

**PROJECT
BUDGETING**

**PROJECT
MANAGEMENT
SOFTWARE**



PM Tools: Software

Low-end

- Basic features, tasks management, charting
- MS Excel

Mid-market

- Handle larger projects, multiple projects, analysis tools
- MS Project (approx. 50% of market)

High-end

- Very large projects, specialized needs, enterprise
- AMS Realtime (Adv Mngt Solution)

overview of the different software project management
tools categorized by their complexity and market usage

Category	Tool	Features	Market Usage
Low-End Tools	MS Excel	Basic task management, simple charting (Gantt charts, timelines), tracking progress	Widely used in small projects or individual task management; ideal for beginners or small teams.
	Trello	Kanban boards, task tracking, basic collaboration tools	Popular for small to medium-sized projects, especially in Agile teams.
	Google Sheets	Similar to MS Excel, with added real-time collaboration features	Used by small teams for simple project tracking, especially when cross-team collaboration is needed.

Mid-Market Tools	MS Project	Gantt charts, resource management, task dependencies, critical path analysis	Common in medium to large projects in industries like construction, software development, and engineering.
	Asana	Task management, project timelines, workload management, integrations with other tools	Popular in small to mid-sized teams, especially in tech and creative industries.
	JIRA	Agile project management, sprint planning, issue tracking, reporting	Widely used in software development for managing Agile projects; suitable for medium to large teams.
	Smartsheet	Project management, collaboration, resource management, reporting	Used in various industries for medium to large projects needing advanced planning and reporting.

High-End Tools	AMS Realtime	Enterprise-level project management, resource allocation, real-time tracking, comprehensive reporting	Used in very large projects with specialized needs, such as in government and aerospace industries.
	Primavera P6 (Oracle)	Enterprise project portfolio management, resource leveling, risk analysis, multi-project management	Preferred in large construction, engineering, and infrastructure projects.
	SAP Project Systems (SAP PS)	Integrated project management within SAP, cost and budget management, resource planning, complex reporting	Used by large enterprises already utilizing SAP for other business processes.
	Planview	Portfolio and resource management, advanced analytics, integration with enterprise systems 	Suitable for large organizations managing a portfolio of projects with advanced analytics and integration.

Project Management Institute

- **PMI stands** for the Project Management Institute, and is a not-for-profit professional membership association for project managers and program managers. **PMI** was started in 1969, and now has a membership of more than 2.9 million professionals around the globe.
- .The Project Management Institute is the organization that gives out the PMP (Project Management Professional) credential, a globally recognized certificate that assures employers that a person is trained and qualified to manage projects.

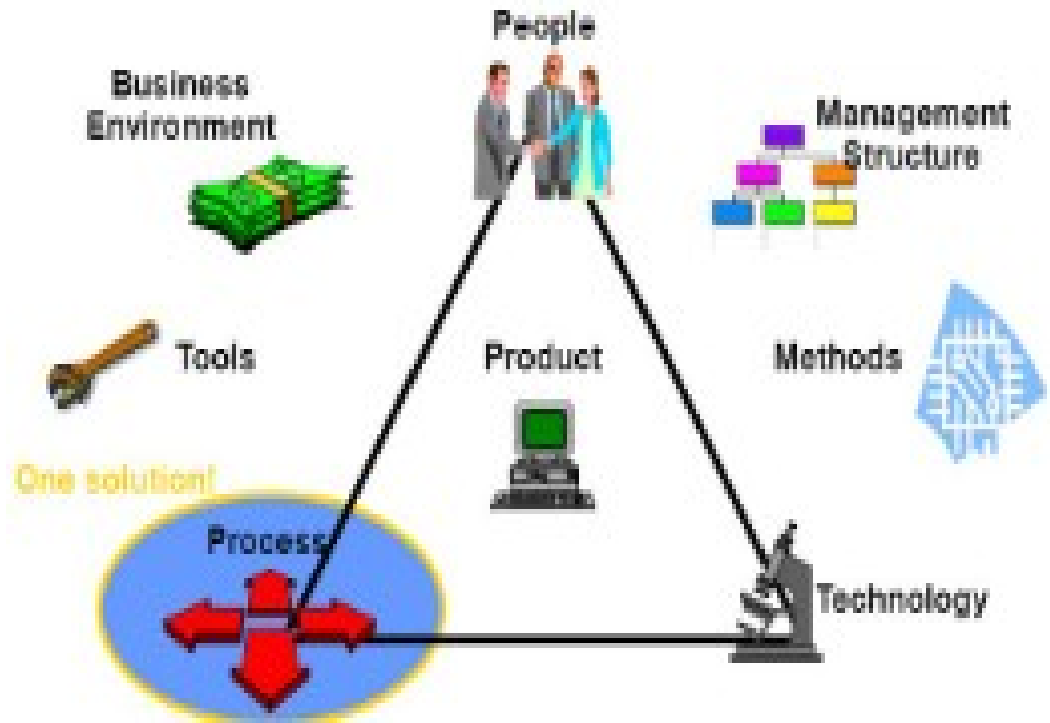
PMI's 10 Knowledge Areas

- Project integration management
- Scope
- Time / Schedule
- Cost
- Quality
- Human resource
- Communications
- Risk
- Procurement
- Stakeholder



Four Project Dimensions

- ☐ People
- ☐ Process
- ☐ Product
- ☐ Technology



Classic Mistakes

- People-Related
- Process-Related
- Product-Related
- Technology-Related

People-Related Mistakes Part 1

- Undermined motivation
- Weak personnel
- Uncontrolled problem employees
- Adding people to a late project

- Noisy, crowded offices
- Customer-Developer friction

■ (**Customer friction** is anything that impedes your **customer** from moving through the marketing and sales funnels to buy your goods or services, or pushes them away from wanting to buy from you again.)

- Unrealistic expectations
- Politics over substance (a particular kind of matter)
- Wishful thinking
- Lack of effective project sponsorship
- Lack of stakeholder buy-in (agreement to do something, even though the person agreeing didn't have anything to do with the project beforehand.)

Process-Related Mistakes Part 1

- Optimistic schedules
- Insufficient risk management
- Contractor failure
- Insufficient planning
- Abandonment of plan under pressure
- Wasted time during fuzzy front end
- Shortchanged upstream activities
- Inadequate design
- Shortchanged quality assurance

- Insufficient management controls
- Omitting necessary tasks from estimates
- Planning to catch-up later
- Code-like-hell programming

Product-Related Mistakes

- Requirements gold-plating (**Gold plating** refers to adding the extra features that were not part of the product scope)
 - Gilding the lily (to attempt to improve something that is already fine the way it is)
- Feature creep (is a tendency for product or project requirements to increase during development beyond those originally foreseen.)
- Developer gold-plating (after having met the requirements, the project manager or the **developer** works on further enhancing the product)
- Push-me, pull-me negotiation (Some managers approve a schedule slip on a project that is progressing more slowly than expected and then add in additional new tasks after the schedule change to make the schedule wrong again)
- Research-oriented development

Technology-Related Mistakes

- Silver-bullet syndrome (is the belief that the next big change in tools, resources or procedures will miraculously or magically solve all of an organization's problems)
- Overestimated savings from new tools and methods
- Switching tools in mid-project
- Lack of automated source-code control

The Five Traditional Process Groups



		Project Management Process Groups				
		Initiating	Planning	Executing	Monitoring & Controlling	Closing
Knowledge Areas	Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work 4.4 Manage Project Knowledge	4.5 Monitor and Control Project Work 4.6 Perform Integrated Change Control	4.7 Close Project or Phase
	Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements 5.3 Define Scope 5.4 Create WBS		5.5 Validate Scope 5.6 Control Scope	
	Project Schedule Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Durations 6.5 Develop Schedule		6.6 Control Schedule	
	Project Cost Management		7.1 Plan Cost Management 7.2 Estimate Costs 7.3 Determine Budget		7.4 Control Costs	
	Project Quality Management		8.1 Plan Quality Management	8.2 Manage Quality	8.3 Control Quality	

Know	Project Resource Management		9.1 Plan Resource Management 9.2 Estimate Activity Resources	9.3 Acquire Resources 9.4 Develop Team 9.5 Manage Team	9.6 Control Resources	
	Project Communications Management		10.1 Plan Communications Management	10.2 Manage Communications	10.3 Monitor Communications	
	Project Risk Management		11.1 Plan Risk Management 11.2 Identify Risks 11.3 Perform Qualitative Risk Analysis 11.4 Perform Quantitative Risk Analysis 11.5 Plan Risk Responses	11.6 Implement Risk Responses	11.7 Monitor Risks	
	Project Procurement Management		12.1 Plan Procurement Management	12.2 Conduct Procurements	12.3 Control Procurements	
	Project Stakeholder Management	13.1 Identify Stakeholders	13.2 Plan Stakeholder Engagement	13.3 Manage Stakeholder Engagement	13.4 Monitor Stakeholder Engagement	

Project Pre-initiation

- It is good practice to lay the groundwork for a project before it officially starts
- Senior managers often perform several pre-initiation tasks, including the following:
 - Determine the scope, time, and cost constraints for the project
 - Identify the project sponsor
 - Select the project manager
 - Develop a business case for a project (see Table 3-2 for an example)
 - Meet with the project manager to review the process and expectations for managing the project
 - Determine if the project should be divided into two or more smaller projects

Project Planning

- The main purpose of project planning is to *guide execution*
- Every knowledge area includes planning information (see Table 3-7 on pages 98-99)
- Key outputs included in the JWD project include:
 - A team contract
 - A project scope statement
 - A work breakdown structure (WBS)
 - A project schedule, in the form of a Gantt chart with all dependencies and resources entered
 - A list of prioritized risks (part of a risk register)
- See sample documents starting on p. 101

Figure 3-4. JWD Consulting Intranet Site Project Baseline Gantt Chart

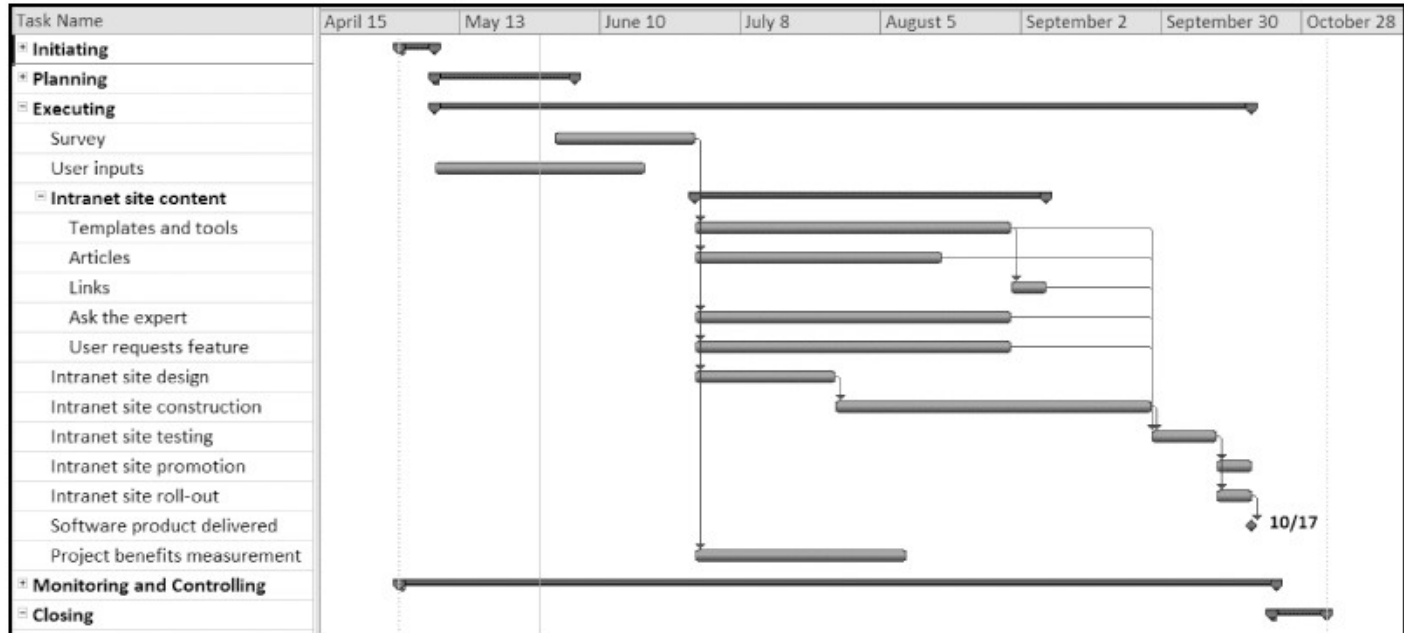


Table. 3-10. List of Prioritized Risks

RANKING	POTENTIAL RISK
1	Lack of inputs from internal consultants
2	Lack of inputs from client representatives
3	Security of new system
4	Outsourcing/purchasing for the article retrieval and “Ask the Expert” features
5	Outsourcing/purchasing for processing online payment transactions
6	Organizing the templates and examples in a useful fashion
7	Providing an efficient search feature
8	Getting good feedback from Michael Chen and other senior consultants
9	Effectively promoting the new system
10	Realizing the benefits of the new system within one year

Project Executing

- Usually takes the most time and resources to perform project execution
- Project managers must use their leadership skills to handle the many challenges that occur during project execution
- Table 3-11 lists the executing processes and outputs. Many project sponsors and customers focus on deliverables related to providing the products, services, or results desired from the project
- A milestone report can help focus on completing major milestones

Part of Milestone Report (Table 3-12, partial)

Milestone	Date	Status	Responsible	Issues/ Comments
<i>Initiating</i> Stakeholders identified	May 2	Completed	Erica and Joe	
Project charter signed	May 10	Completed	Erica	
Project kick-off meeting held	May 13	Completed	Erica	Went very well
<i>Planning</i> Team contract signed	May 13	Completed	Erica	
Scope statement completed	May 27	Completed	Erica	
WBS completed	May 31	Completed	Erica	
List of prioritized risks completed	June 3	Completed	Erica	Reviewed with sponsor and team
Schedule and cost baseline completed	June 13	Completed	Erica	
<i>Executing</i> Survey completed	June 28		Erica	Poor response so far!

Project Monitoring and Controlling

- Involves measuring progress toward project objectives, monitoring deviation from the plan, and taking correction actions
- Affects all other process groups and occurs during all phases of the project life cycle
- Outputs include performance reports, change requests, and updates to various plans
- See Table 3-13

Project Closing

- Involves gaining stakeholder and customer acceptance of the final products and services
- Even if projects are not completed, they should be closed out to learn from the past
- Outputs include project files and lessons-learned reports, part of organizational process assets
- Most projects also include a final report and presentation to the sponsor/senior management

What is a Project Charter in Project Management?

What is a project charter in project management?

A project charter is a formal, typically short document that describes your project in its entirety — including what the objectives are, how it will be carried out, and who the stakeholders are. It is a crucial ingredient in planning out the project because it is used throughout the project lifecycle.

i.e it is a High-level document that authorizes the stating of the project and use of required resources

- Outlines project objectives, deliverables and the required resources required
- Signed and issued by a member of top management of company who takes up role of sponsor
- Not expected to change throughout the life cycle of project
- Project sponsor champions the project, monitors progress of project, provide feedback and help in removing any obstacle
- Serves as a guiding document for all activities concerning the project

The project charter contains the following elements

- Overall objectives of the project and the broad items that are within the scope of the project and those that are out of scope.
- The time schedule in terms of start date and the expected completion date of the project
- Important stakeholders and their responsibilities
- Overview of resources needed and overall budget
- Major risks to the project and strategies for overcoming

The project charter typically documents:

- Reasons for the project
- Objectives and constraints of the project
- Who the main stakeholders are
- Risks identified
- Benefits of the project
- General overview of the budget

How to Create a Project Charter

- Understand project goals and objectives. Identify the project vision and determine the scope of the project
- Define project organization. List all of the essential roles for the project, including customers, stakeholders, and day-to-day project team.
- Create an implementation plan. Outline major milestones, dependencies and timeline for the entire team and stakeholders.
- List potential problem areas. No one wants to be a downer, but adding potential risks and issues to the project charter helps everyone think ahead should the worst happen.

Statement of Work:

The SoW is the document that captures and defines all aspects of your project. You'll note the activities, deliverables and the timetable for the project. It's an extremely detailed document as it will lay the groundwork for the project plan.

WHAT MAKES UP A STATEMENT OF WORK?

1 Introduction

Start with what is being done and who is doing it.

2 Purpose

What is the reason for the project? Explain such in a purpose statement.

3 Scope

What needs to be done in the project, and what process will be used to do it?

4 Where

Note where the project will take place: on site, remotely, etc.

5 Tasks

Break down the scope of the project into smaller tasks that are needed to complete the work.

6 Milestones

Note the phases of the project as milestones to break up the larger schedule.

7 Deliverables

List and explain what is due and when it is due.

8 Schedule

Including due dates of deliverables, add relevant timelines so that it ends with a deadline.

9 Standards & Testing

Note what if any standards are required and if there will be a testing stage to the project.

10 Success

Define what the sponsor and/or stakeholder define as a successful project.

11 Requirements

Determine if there is special equipment necessary for the project or if the team needs certifications.

12 Payments

How will pay be delivered and on what schedule, after milestones, cyclical?

13 Other

Note what didn't fit in previous categories, such as if there are security issues, travel pay, etc.

14 Close

Lastly, list how deliverables will be delivered, as well as signing off and archiving all paperwork.

PROJECTMANAGER

THANK YOU

